

# RAJ

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## PROFESSIONAL SUMMARY

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Data Analyst specializing in high-performance cloud architectures (GCP, AWS, Snowflake) and full-cycle ELT automation to drive enterprise profitability. Proven expertise in engineering scalable data lakehouses and hybrid in-memory pipelines (DuckDB, Parquet, dbt) to process massive datasets (400M+ rows) with zero-maintenance overhead. Strategic business partner with a track record of deploying advanced behavioral models (LTV, Market Basket Analysis) via dynamic BI platforms to eliminate revenue bleed, optimize marketing ROI, and protect profit margins.

## EXPERIENCE

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### Opsie

*Independent Data Consultant*

**July 2025 – April 2026**

*Remote*

- Orchestrated scalable ELT (Extract, Load, Transform) workflows to automate the ingestion of complex, multi-source enterprise datasets into Google Cloud Storage (GCS) for centralized data staging.
- Engineered optimized data architectures in Google BigQuery, utilizing advanced SQL to materialize external data into high-performance native tables, significantly reducing query latency.
- Designed and deployed modular, version-controlled transformation pipelines using dbt Core, ensuring strict data testing, lineage tracking, and reliability.
- Deployed production-ready data marts seamlessly integrated with business intelligence (BI) platforms, eliminating manual overhead and automating real-time analytics reporting.

## PROJECTS

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### Promotional Waste Audit & Margin Optimization | *GCP, BigQuery, DuckDB, dbt, Power BI*

- \* Architected an automated ELT pipeline processing 411M+ rows with multi-table of e-commerce behavioral data, diagnosing a \$2M cart abandonment bleed and optimizing a margin-eroding discount strategy.
- \* Engineered a scalable data lake in Google Cloud Storage utilizing Parquet format for optimized columnar reading, and pioneered a hybrid SQL/Python exploratory workflow with DuckDB in Colab Enterprise to drastically reduce BigQuery compute costs.
- \* Automated complex data transformations and table staging (External to Native) using dbt Core and VS Code, ensuring highly reliable and version-controlled data pipelines.
- \* Deployed dynamic Power BI dashboards utilizing advanced DAX to suppress automatic 10% discounts for Safe Buyers and halt paid retargeting for Window Shoppers, significantly improving ROAS.

### SaaS LTV Optimization & Optimizing Marketing Spend | *AWS S3, Athena, DuckDB, Parquet, dbt, Power BI*

- Engineered an LTV-based acquisition model analyzing 22M+ transaction records and 6M members detail with multi-table for a B2C SaaS platform, identifying severe marketing cash burn and strategic reallocation metrics toward high-retention "Power Users".
- Architected a serverless, decoupled data lakehouse by storing heavy user ledgers in optimized Parquet format within Amazon S3 and querying datasets directly using Amazon Athena to minimize compute overhead
- Pioneered a hybrid DuckDB workflow (SQL + Python) within VS Code to perform lightning-fast, in-memory exploratory data analysis on the 22M row dataset, radically cutting down interactive query latency.
- Automated production-grade transformations and 30-day strict subscriber churn logic by building modular dbt Core pipelines seamlessly integrated with the Athena catalog

### Market Basket Analysis & Cross-Selling Engine | *Snowflake, AWS S3, Tableau, dbt, DuckDB*

- Engineered a cross-selling recommendation engine by performing Market Basket Analysis on 2M+ transactions and 92K products to maximize Average Order Value (AOV).
- Architected a scalable pipeline by dumping Parquet datasets into an AWS S3 data lake and staging them into Snowflake via highly optimized external and native tables.

- Pioneered a hybrid exploratory workflow utilizing DuckDB and Snowflake Notebooks (SQL/Python) to rapidly analyze item combinations and compute conditional probabilities for "Directional Pull" between lead and follow products.
- Automated association rule logic using dbt Core and VS Code to differentiate between low-retention impulse buys and habitual reorders, directly informing high-value "Subscribe & Save" business strategies.

### **Competitive Intelligence & Market Share Analytics** | *DuckDB, Python, SQL, Parquet, Power BI*

- Spearheaded a competitive intelligence initiative analyzing 10M+ rows of retail data to pinpoint revenue bleed, market share erosion, and physical shelf-space loss against primary competitors.
- Engineered a high-performance, zero-infrastructure data pipeline within Google Colab, leveraging Parquet files and a DuckDB hybrid SQL/Python workflow to process massive datasets entirely in-memory.
- Executed geospatial distribution gap analysis across target counties to identify specific cities and retail whitespace where competitor products possessed wider availability.
- Diagnosed store-level product assortment and sales velocity gaps, equipping field sales representatives with actionable, localized intelligence to reclaim lost retail accounts and drive immediate revenue.

## **EDUCATION**

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**Chandigarh University**  
*Bachelor of Engineering*

Sept. 2021 – June 2025  
Mohali, Punjab

## **SKILLS**

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**Languages & Querying:** Python (Pandas), SQL (Advanced), DAX

**Cloud Platforms:** Google Cloud Platform (BigQuery, Cloud Storage), AWS (S3, Athena), Snowflake

**Data Engineering Architecture:** dbt (Data Build Tool), dbt Core, DuckDB (Hybrid), Git/GitHub

**Visualization & Reporting:** Power BI, Tableau, Seaborn, Matplotlib

**Data Notebooks:** Google Colab Enterprise, Snowflake Notebooks, Amazon SageMaker

## **CERTIFICATIONS**

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**Oracle Cloud Infrastructure 2025 Certified Data Science Professional** | *Oracle*

**Google Cloud Data Analytics Certificate** | *Google*